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Developmental differences in linguistic release from masking: evidence from Indian bilingual listeners

Shashank Nema^{1*} , Animesh Barman² and Srikar Vijayasathy²

Abstract

Introduction Improvement in the speech recognition scores due to the language mismatch between target and masker is known as Linguistic release of Masking (LRM). LRM is less explored in children, especially in the Indian scenario.

Aim This study assessed if school going children and adults benefited from a language mismatch between target and masker speech for word recognition in a 2-talker. Effect of various signal to noise ratio (SNR)s on the linguistic release of masking was also studied.

Method Three different age groups (6 to 9years; 10 to 15 years & young adults) each with '15' native Kannada speakers were taken. All the participants were subjected to Kannada PB word recognition task in the presence of two-talker Kannada babble and two-talker Hindi babble at 0 dB SNR, -5 dB SNR and -10 dB SNR. Stimulus was presented at comfortable listening levels of the participants. Linguistic masking release (LRM) was calculated by subtracting percentage of speech identification score (SIS) in the presence of Kannada babble from the SIS in the presence of Hindi babble.

Results LRM scores for the 6 to 9 years children were significantly greater than that of adults and 10-to-15-year individuals at 0 dB SNR and -10 dB SNR. At -5 dB SNR maximum LRM was obtained for participants of 10-to-15-year age group. LRM was highest at -10 dB in all three age groups.

Conclusion Children are able to take advantage of linguistic differences between Kannada and Hindi speech maskers to a greater degree than adults. Lesser exposure of the non-native babble language in children when compared to adults could be one of the reasons for a greater linguistic masking release in children. Linguistic release from masking was the greatest at the poorest SNR. Further, the effects of noise were more deleterious in the younger age group.

Keywords Informational masking, Speech perception in noise, Developmental audiology, Bilingualism, Cross-linguistic perception, Pediatric audiology

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Introduction

Speech perception in noisy environments involves a complex auditory task that poses significant challenges for listeners. Although background noise makes listening difficult for people of all ages however children struggle with it much more. This suggests that the skill of understanding speech in noise develops slowly over time [1]. The difference in speech recognition thresholds between children and adults is more when the background consists of a small number of competing talkers compared to steady-state noise [2]. This indicates that the ability to recognize speech in competing speech, which involves overcoming both energetic and informational masking, matures at a different rate than the ability to cope with steady maskers [3].

The difficulty of listening speech in a speech masker arises from two primary types of masking. First one is energetic masking which occurs when the neural excitation patterns of the target and masker overlap inside the cochlea, making the target signal inaudible. Second one is informational masking (IM), a more central phenomenon, refers to the interference that arises from the similarity, uncertainty, or perceptual confusion between the target and masker, even when the target is audible [4]. IM occurs more when the masker itself is an intelligible speech, as it competes for cognitive resources involved in attention, stream segregation, and linguistic processing [5]. A key factor known to reduce informational masking is the linguistic dissimilarity between the target and the masking signal. Adult listeners with normal hearing consistently demonstrate a “linguistic release from masking” (LRM), where speech recognition improves significantly when the target and masker are spoken in different languages compared to the similar language [6]. This release is due to a reduction in IM, occurred by perceptual dissimilarities at two levels. low-level acoustic cues that initially separate the sound streams, and higher-level cognitive processes that uses the listener’s language skills to focus on the intelligible speech and block out the dissimilar-language background noise [7]. The extent of this benefit depends on the listener’s language experience, such as how early they learned and how proficient they are in the languages involved [8, 9]. However, the current understanding of LRM remains incomplete due to several critical limitations in the existing literature. First, within the linguistically diverse context of India, studies have shown conflicting results. While some research has shown LRM using two-talker babble [10], other studies using multi-talker babble have found no significant effect of the masker’s language, suggesting that energetic masking may overshadow linguistic factors when the number of talkers is high [11]. Interestingly, listeners sometimes perform worse when the distracting speech has a non-native accent, possibly because the unfamiliarity of the

accent itself captures their attention. Calandruccio et al. (2010) further demonstrated that unfamiliar accents may increase listener distraction because they attract attention despite reduced intelligibility, emphasizing that linguistic familiarity and accent familiarity contribute differently to informational masking [12]. These inconsistencies highlight the need to systematically investigate the parameters that govern IM and LRM in these populations.

Second, and more importantly, the literature on LRM is predominantly based on adult listeners. The developmental time course of this ability remains largely unexplored. This is a significant lacuna because the ability to overcome informational masking is closely linked to the maturation of cognitive and linguistic processes. Children, with their less mature auditory processing, smaller working memory capacities, and developing linguistic frameworks, may be less efficient at utilizing the acoustic-phonetic and linguistic cues that facilitate stream segregation and LRM in adults [13]. Understanding this developmental trajectory is essential, as it has direct implications for designing effective auditory rehabilitation strategies and educational accommodations for children in typical classroom settings, which are often characterized by high levels of competing speech. Although previous Indian investigations (e.g., Jain et al., 2014; Anitha, 2023) evaluated native versus non-native masking effects, these studies primarily involved adults and used multi-talker babble. The present study differs by comparing developmental age groups (children, adolescents, and adults) using a two-talker masker, which is known to produce greater informational masking than multi-talker babble. Furthermore, the study examines LRM across multiple SNRs using Kannada and Hindi, two phylogenetically distinct Indian languages. Recent reviews have also highlighted that developmental changes in linguistic release from masking remain poorly understood and warrant systematic investigation in pediatric populations [14]. To our knowledge, this is the first Indian study to investigate developmental differences in linguistic release from masking under these experimental conditions.

Hypotheses

H1 Children in both younger (6–9 years) and older (10–15 years) groups will show significantly lower speech identification scores (SIS) than adults in the presence of both Kannada and Hindi maskers, with the youngest group performing most poorly, particularly at –10 dB SNR.

H2 A significant age $\times$ SNR interaction will emerge, such that the age-related performance gap widens as SNR

decreases (i.e., greater decrement in children at poorer SNRs).

H3 The effect of masker language will be stronger in children than in adults; specifically, the Kannada (same-language, more informative) masker will cause greater performance decrement relative to the Hindi (different-language, less informative) masker in children, reflecting greater informational masking due to immature cognitive-linguistic closure and selective attention skills. Consequently, LRM (the difference between Hindi and Kannada masker conditions) will be larger in children than adults.

H4 A significant three-way interaction (age $\times$ SNR $\times$ masker) is predicted, with the age difference in LRM being largest at the most adverse SNR (-10 dB), where informational masking is most pronounced."

Methods

This study was designed to examine linguistic release from masking across different developmental stages. A cross-sectional design was adopted, and participants were recruited from schools and the community in Mysuru city. The study protocol was ethically approved. And all procedures were performed in accordance with the ethical principles of the Declaration of Helsinki.

Before participation, written informed consent was obtained from all adult participants and from the parents/legal guardians of all child participants. For the minors, their verbal assent was also obtained prior to the commencement of testing. The consent process involved a comprehensive explanation of the study's aims, procedures, potential discomforts, and benefits, with an assurance of data confidentiality and the right to withdraw at any time without consequence.

Participants

Forty-five native Kannada speakers were enrolled and divided into three groups: children (6–9 years, $n = 15$), juveniles (10–15 years, $n = 15$), and young adults (20–35 years, $n = 15$). Children and juveniles were selected from public and government schools of Mysuru, while adults

were recruited through stratified random sampling. (Table 1)

Inclusion criteria

All participants had normal speech-language development, hearing sensitivity within normal limits ($PTA \leq 15$ dB HL, WHO 2008), type A/As tympanograms with acoustic reflexes present (Jerger, 1970), SPIN scores $\geq 60\%$ at 0 dB SNR, and present transient otoacoustic emissions. The 60% cutoff was chosen to ensure participants could perform above chance (25% for 4-alternative forced choice) and to exclude those with possible undiagnosed auditory processing deficits, while acknowledging this may introduce a ceiling effect at 0 dB. Central auditory processing disorder was excluded using SCAP (Vaidyanath & Yathiraj, 2014) for Groups 1–2 and STAP (Yathiraj & Maggu, 2014) for Group 3."

Sample size justification

A formal a priori power analysis was not performed. The sample size of 45 participants (15 per group) was determined based on feasibility and was comparable to previous studies investigating linguistic release from masking. Although statistically significant effects were observed, future studies with larger samples are recommended to improve the precision of interaction estimates and enhance the generalizability of the findings.

Stimuli and instrumentation

Target speech and masker stimuli

Target speech consisted of phonemically balanced Kannada PB word lists (50 words per list, 4 lists total) from the AIISH (All India Institute of Speech and Hearing) database. Each word was a monosyllabic noun (e.g., /mane/ 'house', /ni:ru/ 'water'). Two-talker babble maskers were created by concatenating continuous speech passages from two female native Kannada speakers (ages 32 and 38, Mysore accent) for the Kannada masker, and two female native Hindi speakers (ages 28 and 32, Delhi accent) for the Hindi masker. All speakers were selected based on their native language status and clear articulation. Recordings were made in a sound-attenuated booth (ambient < 30 dB SPL) with a condenser microphone at 44.1 kHz, 16-bit quantization. Maskers were calibrated to match the long-term average speech spectrum of the Kannada PB words.

Masker intelligibility assessment

We did not formally measure masker intelligibility using a speech recognition task, but post-experiment debriefing confirmed that the Kannada babble contained recognizable content words for all participants, whereas the Hindi babble was partially intelligible (participants could identify occasional words but not follow the overall

Table 1 Demographic and audiological characteristics of participant groups

Parameter	Children (6–9 yrs)	Juveniles (10–15 yrs)	Adults (20–35 yrs)
N	15	15	15
Age (years)	7.8 $\pm$ 1.1	12.4 $\pm$ 1.6	26.7 $\pm$ 4.2
Sex (M/F)	7/8	8/7	6/9
PTA (dB HL)	8.7 $\pm$ 3.2	9.5 $\pm$ 2.9	7.8 $\pm$ 2.5
MCL (dB SPL)	61.0 $\pm$ 2.8	63.3 $\pm$ 2.4	65.2 $\pm$ 2.1
SPIN at 0 dB (%)	69.5 $\pm$ 11.7	78.6 $\pm$ 10.2	87.3 $\pm$ 8.4

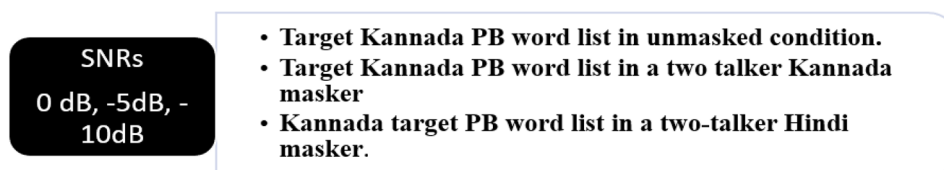


Fig. 1 Experiment details

SIS in the presence of Hindi babble - SIS in the presence of Kannada babble

Fig. 2 Calculation of LRM

meaning). This distinction is critical for interpreting informational masking effects, as the Kannada masker provides greater linguistic interference at both phonetic and semantic levels.

Instrumentation and recording conditions

Pure-tone thresholds and speech audiometry were obtained using a calibrated two-channel diagnostic audiometer (GSI-61) with TDH-50 headphones, MX/41AR ear cushions, and a RadioEar B-71 bone vibrator. Middle-ear status and acoustic reflexes were assessed using a GSI TymStar (Grason-Stadler, USA). Transient otoacoustic emissions were measured with an ILO V-6 system. Central auditory processing was screened using the SCAP [15] and STAP [16] tools.

Stimulus preparation was performed on a Lenovo G-50 laptop with Adobe Audition v3 and a MOTU MicroBook II audio interface. The target speech and babble were first RMS-normalized to 50% amplitude and then digitally mixed at specific SNRs using a custom script. The masking release experiment was administered via Sennheiser HDA 200 circum-aural headphones connected through an MX-150 audio jack. Testing was conducted in an air-conditioned, sound-treated room with ambient noise maintained within permissible limits [17].

Linguistic Release from Masking (LRM) experiment

Following confirmation of normal auditory function, participants were seated in a sound-treated room and instructed to repeat target Kannada PB words while ignoring competing babble. Five practice trials at 0 dB SNR were administered to familiarize participants with the task.

Mode and level of presentation

Target PB words were presented diotically through a calibrated Sennheiser HDA 200 headphone connected to a Lenovo G-50 laptop. Diotic presentation was used to control for head-shadow effects and binaural release from masking, which could confound age comparisons. While this approach provides a controlled laboratory

measure of informational masking, it does not reflect natural binaural listening conditions.

Three masker conditions were tested at 0, -5, and -10 dB SNR: (i) unmasked target, (ii) two-talker Kannada masker, and (iii) two-talker Hindi masker (Fig. 1). Stimuli were delivered at each participant's most comfortable listening level (MCL). MCL was established by presenting the PB word lists at increasing levels (starting at 40 dB SPL) and asking participants to select the level they found 'most comfortable for understanding speech.' Absolute MCL values were: adults 65.2 ± 2.1 dB SPL (range 62–68); juveniles (10–15 years) 63.3 ± 2.4 dB SPL (60–66); children(6years) 61.0 ± 2.8 dB SPL (58–64). SNR was then created by fixing the speech level at MCL and adjusting the masker level accordingly (e.g., for -10 dB SNR, masker = MCL + 10 dB).

Scoring and analysis

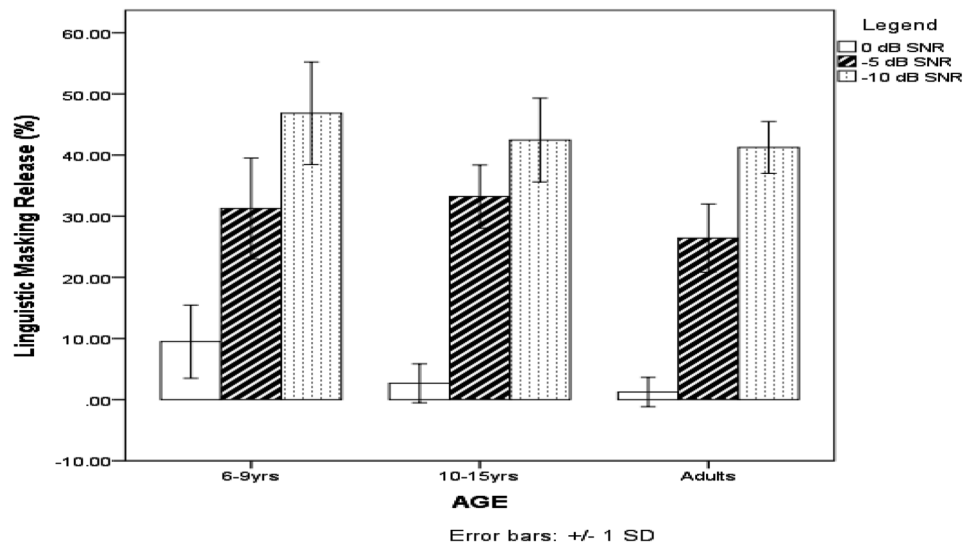
SIS scores were calculated as the percentage of correctly repeated words (%SIS = [correct responses ÷ total words] × 100). Linguistic Release from Masking (LRM) was calculated as the difference between the Speech Identification Scores (SIS) obtained in the Hindi masker and Kannada masker conditions (LRM = SIS in Hindi masker - SIS in Kannada masker), as illustrated in (Fig. 2). This difference represented the improvement in speech recognition attributable to the language mismatch between the target and masker.

Results

The study investigated the effect of non-native maskers on native language target identification, comparing linguistic release of masking (LRM) across age groups and SNR conditions (0, -5, -10 dB). Participants completed the Kannada PB word recognition task with two-talker Kannada and Hindi maskers. LRM scores for all groups were entered into SPSS v21. Shapiro-Wilk tests indicated that scores at -5 dB and -10 dB SNR were normally distributed across all groups, whereas scores at 0 dB SNR were skewed. Inspection of score distributions showed clustering of observations near the maximum possible score at 0 dB SNR, suggesting ceiling effects. These

Table 2 Z values along with level of significance across SNR and age for SIS obtained between two maskers

SNR	0 dB SNR		-5 dB SNR		-10 dB SNR	
Language/ age	Z	P	Z	p	Z	p
6 to 9 years	-1.51	0.069	-3.62	0.000	-3.62	0.00
10 to 15 years	-1.81	0.099	-3.525	0.000	-3.52	0.00
Adults	-1.89	0.069	-3.54	0.000	-3.54	0.00

**Fig. 3** Depicting mean and SD of linguistic release of masking at different SNR across the age groups

results guided the selection of parametric or non-parametric statistical analyses for subsequent comparisons.

Effect of native and non-native masker on speech identification

Analysis of the results revealed that the linguistic similarity of the masker to the target signal significantly impacted speech identification scores. A Friedman test confirmed a significant difference in SIS between Kannada and Hindi maskers across SNRs in all age groups [χ^2 [6] = 248.29, $p < 0.05$]. Pairwise comparisons using the Wilcoxon signed-rank test demonstrated that SIS was significantly poorer in the presence of Kannada masker compared to Hindi masker at -5 dB and -10 dB SNR across all groups (6–9 yrs: $Z = -3.62$, $p < 0.001$; 10–15 yrs: $Z = -3.52$, $p < 0.001$; Adults: $Z = -3.54$, $p < 0.001$). At 0 dB SNR, differences were not statistically significant ($p > 0.05$) (Table 2).

Linguistic Release of Masking (LRM) across age and SNR

The magnitude of Linguistic Release from Masking (LRM) was quantified as the difference between speech identification scores obtained in the Hindi and Kannada masker conditions and compared across age groups and SNRs. A Two-way mixed-design ANOVA (mixed repeated-measures ANOVA) revealed a significant main effect of age group, $F(2,45) = 9.336$, $p < 0.001$, **partial $\eta^2 = 0.293$** , indicating that LRM differed significantly across

age groups. There was also a significant main effect of SNR, $F(2,90) = 695.217$, $p < 0.001$, **partial $\eta^2 = 0.939$** , demonstrating that LRM varied substantially across SNR conditions. Furthermore, a significant Age Group $\times$ SNR interaction was observed, $F(4,90) = 3.297$, $p = 0.014$, **partial $\eta^2 = 0.128$** , indicating that the effect of SNR on LRM differed across age groups. (Fig. 3) Bonferroni-adjusted post hoc comparisons revealed that children aged 6–9 years demonstrated significantly greater LRM than adults ($p < 0.05$), whereas the 10–15-year group did not differ significantly from either of the other age groups.

At 0 dB SNR, data were non-normal; therefore, Kruskal–Wallis analysis was conducted, revealing a significant difference in LRM across groups [χ^2 [2] = 21.53, $p < 0.05$]. Post-hoc Mann–Whitney U tests indicated that 6–9-year-olds had significantly greater LRM than both 10–15-year-olds ($Z = -3.42$, $p < 0.001$) and adults ($Z = -4.16$, $p < 0.001$), whereas no significant difference was observed between the latter two groups ($p > 0.05$).

Effect of age on LRM within SNR

The magnitude of the Linguistic Release from Masking (LRM) was quantified and compared across the different age groups and SNR conditions. Independent t -tests showed that at -5 dB SNR, LRM scores of 10–15-year-olds were significantly higher than adults ($t = 3.59$, $p < 0.001$), while no other group differences reached significance. At -10 dB SNR, LRM was significantly higher

Table 3 t- values along with significant level LRM obtained between the age groups at '-5'dB&'10' dB SNR

SNR	6 to 9 v/s 10 to 15		10 to 15 v/s Adults		Adults v/s 6 to 9	
	t	p	t	p	t	p
'5'dB	-0.8	0.48	3.59	0.00	1.95	0.06
-10 dB	1.61	0.11	0.59	0.00	2.38	0.02

Table 4 Z values along with significant levels obtained at 0 dB SNR between the age groups

Age group	t- values	p-value
6 to 9 years	-7.29	0.000
10 to 15 years	-6.36	0.000
Adults	-11.10	0.000

Table 5 t- values along with the significant levels between - '5'dB and '-10' dB SNR within each age group

At 0 dB SNR	6 to 9 v/s 10 to 15	10 to 15 v/s Adults	Adults v/s 6 to 9
Z- value	-3.42	-1.68	-4.16
p- Value	0.00	0.93	0.00

in 6–9-year-olds compared to adults ($t=2.38$, $p=0.02$), with no significant differences between the other groups. (Table 3)

Effect of SNR on LRM within age groups

Within all age groups, the Linguistic Release from Masking showed significant variation across different signal-to-noise ratios. Paired-sample t-tests revealed that in all age groups, LRM scores were significantly higher at -10 dB compared to -5 dB (6–9 yrs: $t = -7.29$, $p < 0.001$; 10–15 yrs: $t = -6.36$, $p < 0.001$; Adults: $t = -11.10$, $p < 0.001$) (Table 4). For the skewed 0 dB SNR data, Wilcoxon signed-rank tests confirmed that LRM was significantly lower at this SNR compared to both -5 dB and -10 dB across all age groups (Table 5) $Z > -3.5$, all $p < 0.001$).

Discussion

The present study investigated child–adult differences in linguistic release of masking (LRM) during speech-in-speech recognition using Kannada words in the presence of two-talker native (Kannada) and non-native (Hindi) babble at different SNRs. Results showed significantly better recognition in Hindi babble than in Kannada babble across SNRs, except at 0 dB where ceiling effects were present. This finding is similar to the previous researches showing that linguistic mismatch reduces informational masking [8, 9]. Contrastingly, studies using multi-talker babble did not observe LRM [10, 11], possibly due to increased energetic masking when more talkers were present. In the present study, the linguistic dissimilarity between Kannada (Dravidian) and Hindi (Indo-Aryan) may have further enhanced the release.

Energetic masking factors, such as fundamental frequency (F0) and long-term average speech spectrum (LTASS), were also considered but did not fully explain the differences, similar to the previous studies [8, 18]. Thus, the findings largely support linguistic mismatch as the primary contributor to LRM.

“Effect of age was seen in the form of greater LRM for younger children (6–9 years) than adults and older children at -10 dB SNR. Although children performed poorly overall in noise, the larger gap between masked and released conditions enhanced LRM. These results are consistent with the findings of previous research which revealed that immature cognitive control and reduced inhibitory capacity are responsible for children's poorer speech-in-noise performance (Hall et al., 2002; Leibold, 2017).

However, we caution against attributing these differences solely to reduced exposure to Hindi, as language exposure was not directly quantified in the present study. Instead, we interpret our findings as reflecting the combined influence of (a) developing linguistic knowledge (lexical, phonological, and syntactic), (b) immature executive functions including inhibitory control and selective attention, and (c) less efficient stream segregation abilities in children (Wightman & Kistler, 2005). These cognitive-linguistic factors likely interact with language exposure, but disentangling their relative contributions requires future studies incorporating standardized language proficiency measures (e.g., receptive vocabulary, verbal fluency, language history questionnaires) and cognitive assessments (e.g., working memory, attention).”

Signal to noise ratio was a significant factor in LRM, with greater release at poorer SNRs. At better SNRs, better audibility of target reduces both energetic and informational masking [18] while at poor SNRs informational masking was more pronounced, thereby enhancing the benefit of linguistic mismatch [6].

Limitations and future directions

Several limitations of the present study should be acknowledged. First, our sample was drawn from a single urban region (Mysuru city), limiting socioeconomic and educational diversity. Second, diotic presentation does not reflect natural binaural listening, wherein head-shadow and binaural squelch can reduce informational masking; our results may therefore not fully generalize to real-world listening environments. Third, our inclusion criterion (SPIN $\geq 60\%$ at 0 dB) may have selected

a higher-performing subset, and although we applied non-parametric statistics to handle ceiling effects at 0 dB, replication with unselected samples is warranted. Fourth, our findings apply specifically to Kannada-Hindi bilingual contexts and may not generalize to other Indian language pairs or to bilingual populations with different proficiency levels. Fifth, we did not measure working memory, attention, or language proficiency directly, which could have mediated the age $\times$ masker interaction. Given the well-established role of executive functions in speech-in-noise perception [19].

Future studies should incorporate cognitive batteries (e.g., working memory, inhibitory control), binaural listening conditions, and more diverse populations to extend external validity. Additionally, the use of sentence-level stimuli, larger randomized samples, and direct measures of language exposure and proficiency would strengthen the generalizability of these findings."

Conclusion

In conclusion, this study provides evidence that both children and adults benefit from target-masker language mismatch in speech recognition tasks, with younger children (6–9 years) showing greater LRM. Children's larger release may be attributed to higher vulnerability to babble and cognitive immaturity [19]; however, these interpretations require direct verification with language exposure and cognitive measures as language exposure was not directly measured in this study. Furthermore, larger LRM at poor SNRs highlights the interplay between energetic and informational masking. These findings contribute novel evidence from Indian languages and underscore the need for further research using sentence-level stimuli, cognitive assessments, and larger, randomized samples to confirm and extend these observations. Importantly, the generalizability of our findings is constrained by our specific stimulus conditions (diotic presentation, two-talker babble), sample characteristics (urban, Kannada-Hindi context), and inclusion criteria, as detailed in the Limitations section.

Abbreviations

LRM	Linguistic Release from Masking (or Linguistic Release of Masking)
SCAP	Screening Checklist for Auditory Processing
SNR	Signal-to-Noise Ratio
SPIN	Speech Perception in Noise
SPSS	Statistical Package for the Social Sciences
STAP	Screening Test for Auditory Processing
SIS	Speech Identification Score

Author contributions

Shashank Nema has collected data and prepared the manuscript, Animesh Barman has conceptualized the research guided it also performed review and editing of the manuscript. Srikar Vijayarathy has contributed in Statistics and stimulus preparation.

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The study was not funded by any source.

Data availability

Data generated and analyzed for this study (except for audio/video recordings) are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Informed consent was obtained from all participants involved in the study. They were provided with detailed information about the study's purpose, procedures, and potential risks, and they voluntarily agreed to participate by signing a consent form. Measures were implemented to ensure confidentiality and anonymity, protecting the privacy of the participants, and ensuring that their identities remain undisclosed. Since the study was a part of dissertation, ethical committee of research formed in the All-India Institute of Speech & Hearing, Mysore declared the study ethical. Before participation, written informed consent was obtained from all adult participants and from the parents/legal guardians of all child participants. For the minors, their verbal assent was also obtained prior to the commencement of testing. The consent process involved a comprehensive explanation of the study's aims, procedures, potential discomforts, and benefits, with an assurance of data confidentiality and the right to withdraw at any time without consequence.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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